

Investigating the Effects of Limited Field of View and Resolution in Optical See-Through Augmented Reality in the Context of Immersive Analytics

Julia Hertel*

Jan Nicolai Synwoldt†

Frank Steinicke‡

Human-Computer Interaction, University of Hamburg

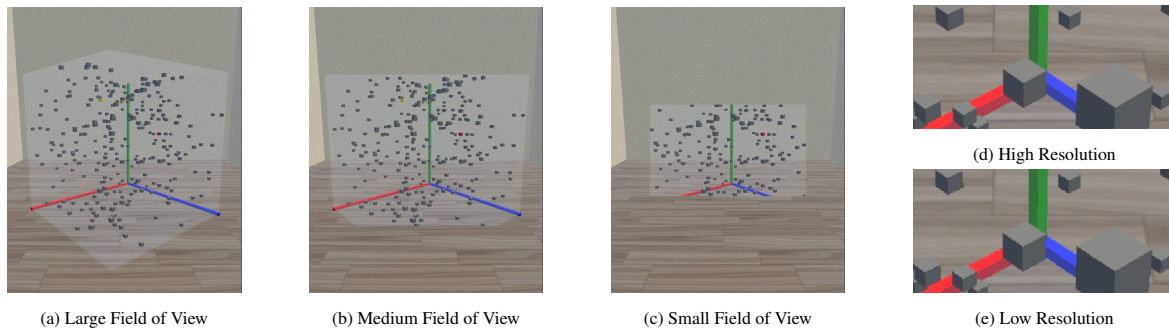


Figure 1: Visualization of the display configurations used in the user study for different *field of views* ((a) - (c)) and *resolutions* ((d) - (e)). The image size of figures (a) - (c) depict the size of the largest field of view. For the medium and small field of view, the size corresponds to the visible area of the virtual content. For the resolution (d) - (e), the difference is especially noticeable on the data points' edges.

ABSTRACT

Currently, optical see-through augmented reality HMDs (OST AR HMDs) still have significant display limitations. This work presents a user study conducted with an OST AR HMD to investigate how two of these limitations - a small field of view and low resolution - affect task performance, user experience, workload, and simulator sickness in the context of immersive analytics. The Magic Leap 2 was used to simulate the display properties of other commonly used HMDs. The results suggest that a limited field of view negatively impacts user experience and workload, with the difference between the Magic Leap 2's field of view and that of the two generations of the Microsoft HoloLens being particularly striking.

Index Terms: Human-centered computing, visualization, mixed / augmented reality.

1 INTRODUCTION

Currently available commercial optical see-through augmented reality head-mounted displays (OST AR HMDs) have noticeable display limitations that might negatively impact users. For instance, reducing the field of view (FoV) of the HoloLens (1st generation), increases task completion times for search tasks [7]. Also, increasing the FoV beyond current HMDs capabilities, e.g., by large external screens, can reduce task completion times [5]. Besides task performance, observations and user feedback from other studies suggest that the limitations of today's HMDs could affect other metrics, such as user experience [2], comfort [6], and required head movements [4]. Also, the effects of other display limitations are investigated, such as resolution, latency, frame rate and jitter [3].

*e-mail: julia.hertel@uni-hamburg.de

†e-mail: j.synwoldt@gmail.com

‡e-mail: frank.steinicke@uni-hamburg.de

This study explores the effect of *FoV* and *resolution*, two common limitations of current AR HMDs, by investigating the hypotheses that limiting display properties decreases the task performance (**H1**) and user experience (**H2**) and increases the perceived workload (**H3**) and simulator sickness (**H4**).

2 METHODS

To compare different display properties, the same HMD was used for all conditions, on which the limited display properties were simulated, to avoid confounding variables that could arise from using different HMDs. The experimental application was developed for the Magic Leap 2, using the Unity engine. Three levels of FoV (horizontal x vertical) were used, matching three common HMDs: $45^\circ \times 55^\circ$ (the unrestricted FoV of the Magic Leap 2), $43^\circ \times 29^\circ$ (HoloLens 2) and $30^\circ \times 17.5^\circ$ (HoloLens 1). For resolution, the unrestricted angular resolution of the Magic Leap 2 was used (32 PPD), as well as the one of the Meta Quest 2 (20 PPD).

As experimental tasks, the immersive analytic tasks presented by Bach et al. [1] were used. For each task, a 3D scatter plot with an edge length of 25 cm was displayed at eye level, which the participants could interact with using a controller. For the Distance task, the diagram showed a random distribution with two pairs of points highlighted in color. Participants had to determine the pair for which the points were closest. For the Cluster task, points were shown in clusters, and participants had to determine the smallest number of clusters that can be seen from any direction. For the Selection task, a random distribution with four highlighted points was shown. Participants had to select the highlighted points using the controller. For the Cuttingplane task, points were shown in three clusters, and participants had to place a plane so that it cuts through the centroids of all the clusters.

3 USER STUDY

The study was conducted as a within-participant design with six conditions (3 FoV x 2 resolution). For each condition, all four tasks were performed once. The condition order was counterbalanced. Before starting the experimental tasks, participants signed

an informed consent, completed a demographic questionnaire, and performed each task once as a trial. After each condition, they completed a questionnaire that included the Simulator Sickness Questionnaire (SSQ), the raw NASA Task Load Index (NASA TLX), and the short version of the User Experience Questionnaire (UEQ-S). The task performance was assessed by measuring the task completion time and the error rate. Depending on the task, the error rate was assessed as the rate of incorrect answers (Distance and Cluster), the rate of incorrect selections (Selection), or the error distance (Cuttingplane). Data was collected from 41 participants (21 women, 19 men and one non-binary person). 19 participants were aged 18-24, 14 were aged 25-34, 3 were aged 35-44, 3 were aged 55-64, one was aged 45-54, and one was aged 65 or older.

According to the rules of the ethics commission of the Department of Informatics of the University of Hamburg, a basic questionnaire was completed, and based on the results, consultation with the ethics commission for approval was not required.

4 RESULTS

For each metric, a two-way repeated measures ANOVA was performed to investigate the influence of *FoV* and *resolution* on the metric, treating participant ID as a within-subject factor. For the task completion time, the data was transformed using a log-10-transformation to reach a normal distribution. An ANOVA with Greenhouse-Geisser corrections revealed a significant main effect of *FoV* for Distance ($F = (1.97, 78.85) = 6.21, p < 0.01$), Cluster ($F = (1.95, 78) = 3.52, p = 0.04$), and Selection ($F = (1.76, 70.52) = 10.18, p < 0.01$). However, Tukey post-hoc tests did not show any significant contrasts. For the error rate, the Aligned Rank Transform (ART) procedure was first applied, followed by an ANOVA. For the Distance task, a significant interaction effect between *FoV* and *resolution* ($F(2, 200) = 8.51, p < 0.01$) and a significant main effect of *resolution* ($F(1, 200) = 4.42, p = 0.04$) were found. A Tukey post-hoc test found no significant contrasts for the interaction effect. For the main effect, a Tukey post-hoc test indicated a smaller error rate for a high *resolution* compared to a low *resolution* ($p = 0.04$). For the Cluster task and the Selection task, no significant interaction or main effects were found. For the Cuttingplane task, a significant main effect of *FoV* ($F(2, 200) = 4.86, p < 0.01$) was found. A Tukey post-hoc test indicated a lower error rate for a large *FoV* compared to both a medium *FoV* ($p = 0.02$) and a small *FoV* ($p = 0.02$). Hence, **H1** can only be accepted partially. For the user experience, the ART was first applied, followed by an ANOVA. For the overall user experience, a significant main effect of *FoV* was found ($F(2, 200) = 11.59, p < 0.01$). A Tukey post-hoc test indicated a higher overall user experience for a large *FoV* compared to a medium *FoV* ($p = 0.04$) and a small *FoV* ($p < 0.01$). For the pragmatic quality, a significant main effect of *FoV* was found ($F(2, 200) = 31.27, p < 0.01$). A Tukey post-hoc test indicated a higher pragmatic quality for a large *FoV* compared to a medium *FoV* ($p < 0.01$) and a small *FoV* ($p < 0.01$), as well as for a medium *FoV* compared to a small *FoV* ($p < 0.01$). For the hedonic quality, no significant interaction effects or main effects were found, leading to a partial acceptance of **H2**. For the perceived workload, the ART was first applied, followed by an ANOVA. For the mental workload, a significant main effect of *FoV* was found ($(F = 2, 200) = 9.05, p < 0.01$). A Tukey post-hoc test indicated a lower mental workload for a large *FoV* compared to a medium *FoV* ($p < 0.01$) and a small *FoV* ($p < 0.01$). For the physical workload, a significant main effect of *FoV* was found ($(F = 2, 200) = 7.64, p < 0.01$). A Tukey post-hoc test shows a lower physical workload for a large *FoV* compared to a medium *FoV* ($p < 0.01$) and a small *FoV* ($p < 0.01$). For the temporal workload, no significant interaction effects or main effects were found. For the performance, a significant main effect of *FoV* was found ($(F = 2, 200) = 3.59, p = 0.03$). A Tukey post-hoc test indicated

a higher performance for a large *FoV* compared to a small *FoV* ($p = 0.02$). For the effort, no significant interaction effects or main effects were found. For the frustration, a significant main effect of *FoV* was found ($(F = 2, 200) = 5.03, p < 0.01$). A Tukey post-hoc test indicated a lower frustration for a large *FoV* compared to a small *FoV* ($p < 0.01$). For the total workload, a significant main effect of *FoV* was found ($(F = 2, 200) = 6.07, p < 0.01$). A Tukey post-hoc test indicated a lower total workload for a large *FoV* compared to a small *FoV* ($p < 0.01$). Since no effects for the resolution were found, **H3** can only be accepted partially. For the simulator sickness, the ART was first applied, followed by an ANOVA. For each scale, no significant interaction effects or main effects were found, thus **H4** cannot be accepted.

5 DISCUSSION AND CONCLUSION

The results show that in particular the *FoV* affects participants, mainly subjectively. They perceived lower task performance, were more frustrated, and reported a higher mental and physical workload. While previously effects of *FoV* on task completion times were found [7, 5], this study could not show such results. This might be due to the use of different experimental tasks. For instance, while in these studies virtual objects had to be found across a large area around the participants, here they interacted with a comparably small plot in front of them. Furthermore, hardly any effects of resolution could be found, although this could also be due to the chosen tasks or the investigation of too low resolution values. Interestingly, only a few significant effects were found between the *FoV* of both generations of the HoloLens, while both *FoVs* differ mainly in comparison to that of the Magic Leap 2, suggesting a complex influence of the size and possibly the shape of the *FoV*. Overall, this study indicates that the subjective experience of users could be influenced by the choice of HMD, which should be carefully considered in further studies and applications.

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